

Lab Assignment #01

Aim:

Model a WR-137 rectangular waveguide with air as its medium and simulate its electromagnetic behavior using CST Studio Suite, with the aim of verifying dominant TE₁₀ mode propagation and analyzing key waveguide performance parameters (modal characteristics, field distributions, scattering parameters, power flow and surface current distributions).

Theory:

A **rectangular waveguide** is a hollow metallic structure with a rectangular cross-section used to transport electromagnetic energy at high frequencies (typically microwave and millimeter-wave bands). Unlike coaxial cables which use a central conductor, waveguides propagate waves using the principle of **total internal reflection**. The electromagnetic waves bounce in a "zigzag" pattern off the inner conducting walls.

The waveguide acts as a **High-Pass Filter**. It allows frequencies above a specific **Cutoff Frequency (f_c)** to propagate with minimal attenuation, while frequencies below this threshold are attenuated.

Waveguides support multiple modes of propagation (TE, TM). The mode with the lowest cutoff frequency is called the **Dominant Mode**. For a standard rectangular waveguide where the width a is larger than height b ($a > b$), the dominant mode is **TE₁₀**.

For TE modes Electric field (E) is entirely transverse to the direction of propagation (no E -component in the z -direction), while the Magnetic field (H) has a component in the direction of propagation.

Table 1 - Parametric Details of WR-137

Parameter	Value (Millimeters)
Broad Dimension (Width)	34.8488 mm
Narrow Dimension (Height)	15.7988 mm
Cutoff Frequency of Lowest Order Mode:	4.301 GHz
Cutoff Frequency of Upper Mode:	8.603 GHz
Recommended Frequency Band:	5.85 to 8.20 GHz
Length	100 mm (suggested)
Wall Material	PEC
Core Material	Air

Procedure:

1. Open CST Studio Suite and select the New Template option from the main dashboard.
2. Select the Microwave & RF / Optical template folder, choose Waveguides & Components, and then select Waveguides.
3. Select the Time Domain Solver from the available solver options and proceed to the next step.
4. Configure the project units by setting Dimensions to mm and Frequency to GHz.
5. Define the Frequency Range with a Minimum of 4.301 GHz and a Maximum of 8.603 GHz to cover the cutoff and operating bands for standard WR-137.
6. Open the Modeling tab, select Background, and set the Material Type to PEC (Perfect Electric Conductor) with zero spacing in all directions.
7. Open the Parameter List window and create a new parameter named a with a value of 34.8488 to represent the broad dimension.
8. Create a second parameter named b with a value of 15.7988 to represent the narrow dimension.
9. Create a third parameter named c with a value of 100 to represent the length of the waveguide.
10. Select the Brick tool from the Modeling tab and press Esc to open the numerical entry dialog box.
11. Configure the brick dimensions using the parameters with Xmin as $-a/2$, Xmax as $a/2$, Ymin as $-b/2$, Ymax as $b/2$, Zmin as $-c/2$, and Zmax as $c/2$.
12. Assign the material for the brick as Air and click OK to generate the geometry.
13. Activate the Pick Face tool (shortcut F) and double-click the front face of the waveguide brick located at $Z=-c/2$.
14. Navigate to the Simulation tab, click on Waveguide Port, and click OK to define Port 1.
15. Activate the Pick Face tool again and double-click the rear face of the waveguide brick located at $Z=c/2$.
16. Navigate to the Simulation tab, click on Waveguide Port, and click OK to define Port 2.
17. Click on Setup Solver in the Simulation tab and ensure the Normalized to 50 Ohm option is unchecked.
18. Click Start to initialize the Time Domain solver and wait for the simulation to complete.
19. Navigate to the 1D Results folder in the navigation tree to retrieve the S-Parameter plots.
20. Navigate to the 2D/3D Results folder to visualize and export the field distributions for E-Field, H-Field, Surface Current, and Power Flow at the monitored frequencies.

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3D Model:

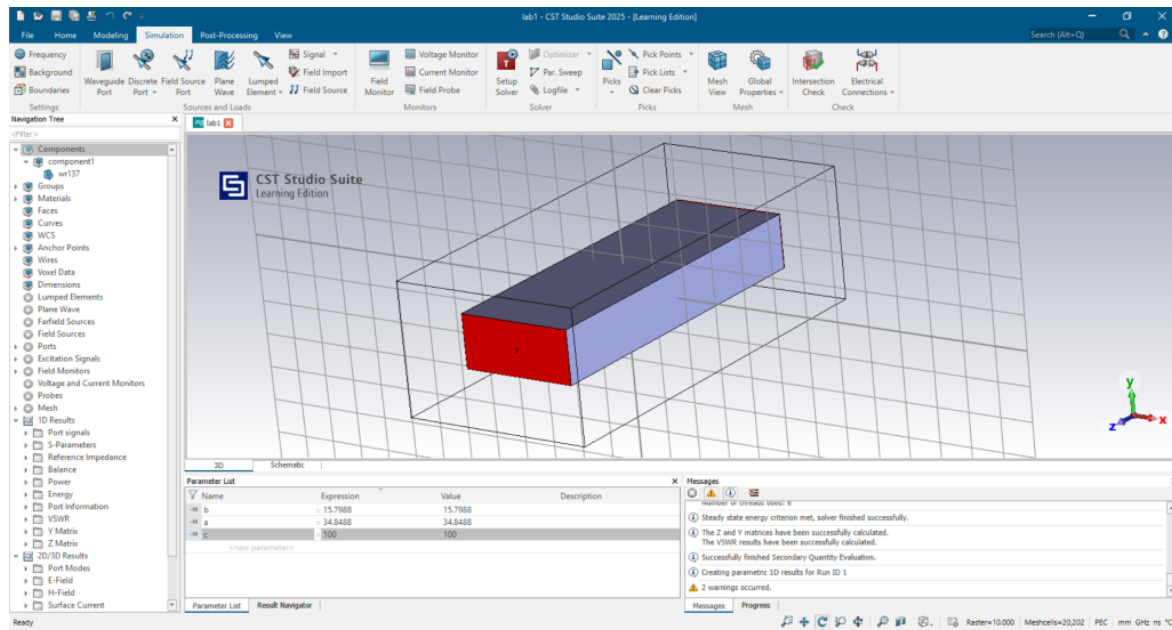


Fig 1.1 3D Model Port 1 View

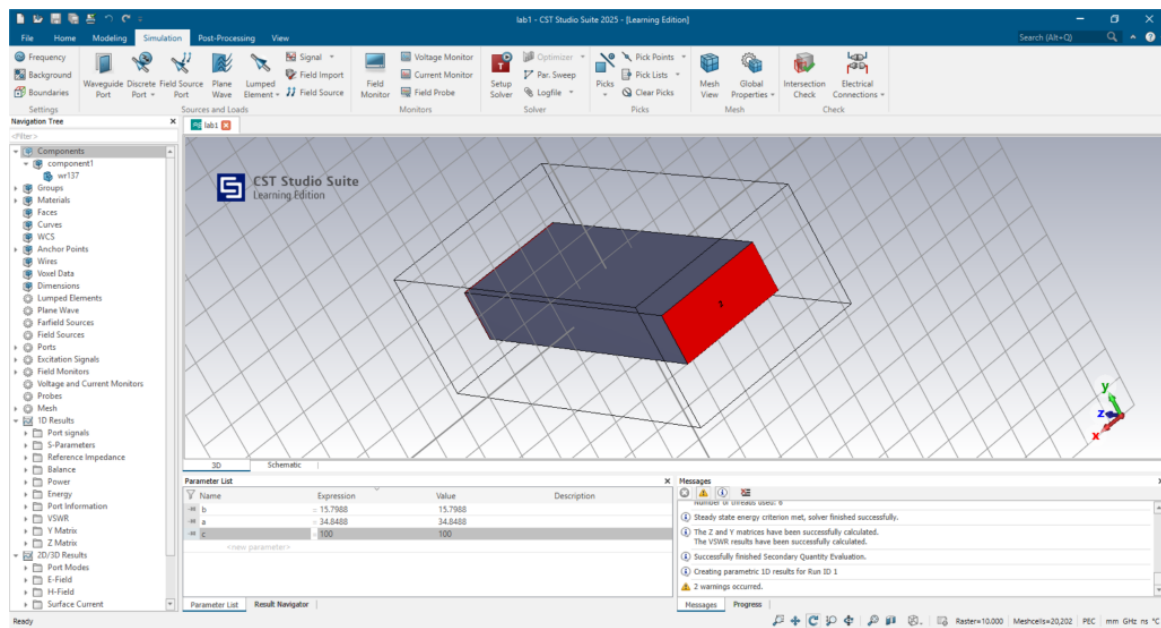


Fig 1.2 3D Model Port 2 View

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Results:

Below are the results for simulating EM waves through a WR-137 rectangular waveguide with air as a medium, over the range of frequencies 4.301 GHz to 8.603 GHz.

Electric Field Distributions

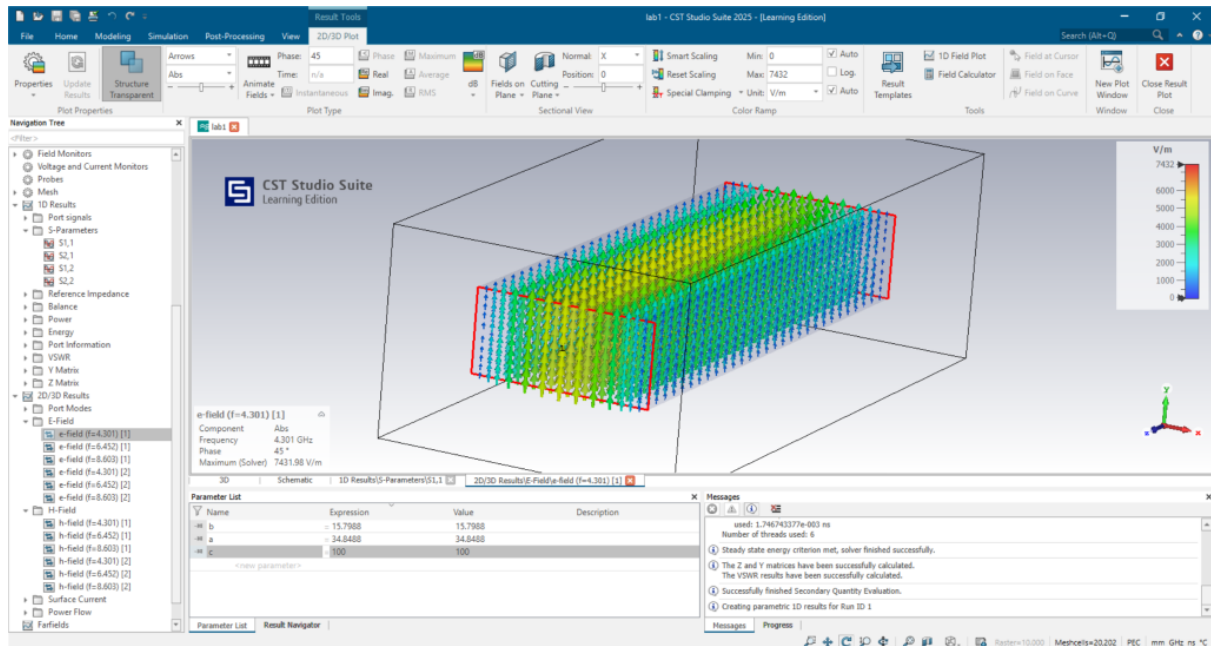


Fig 2.1 Electric field distribution for frequency = 4.301 GHz at Port 1

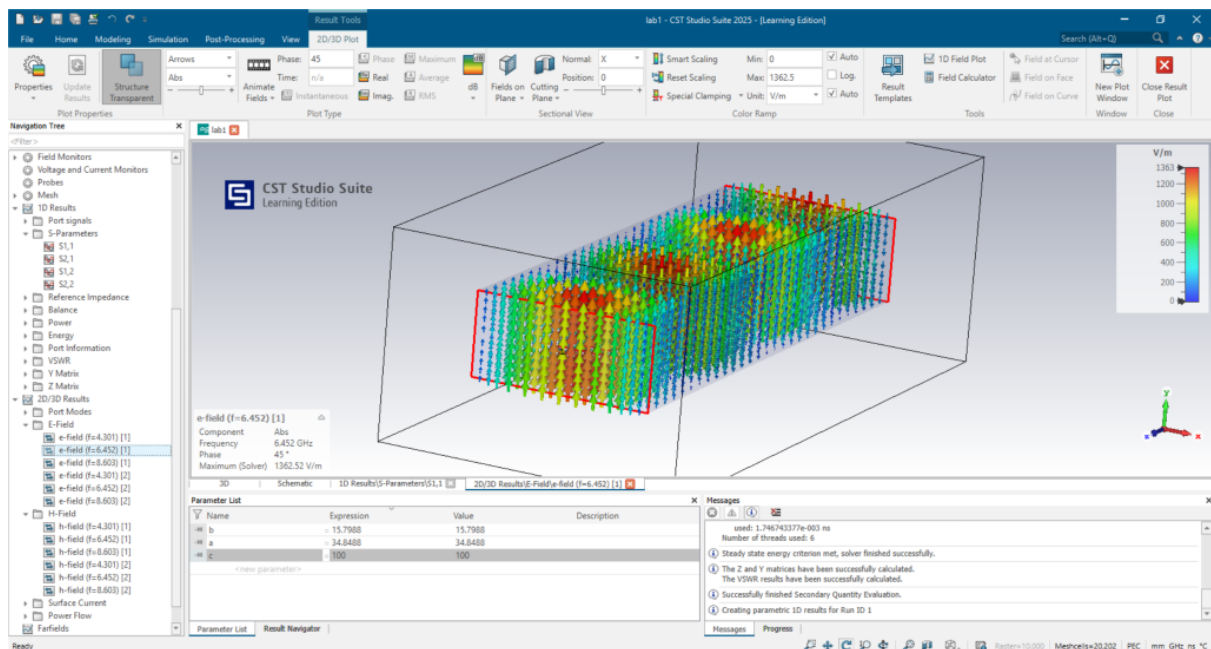


Fig 2.2 Electric field distribution for frequency = 6.452 GHz at Port 1

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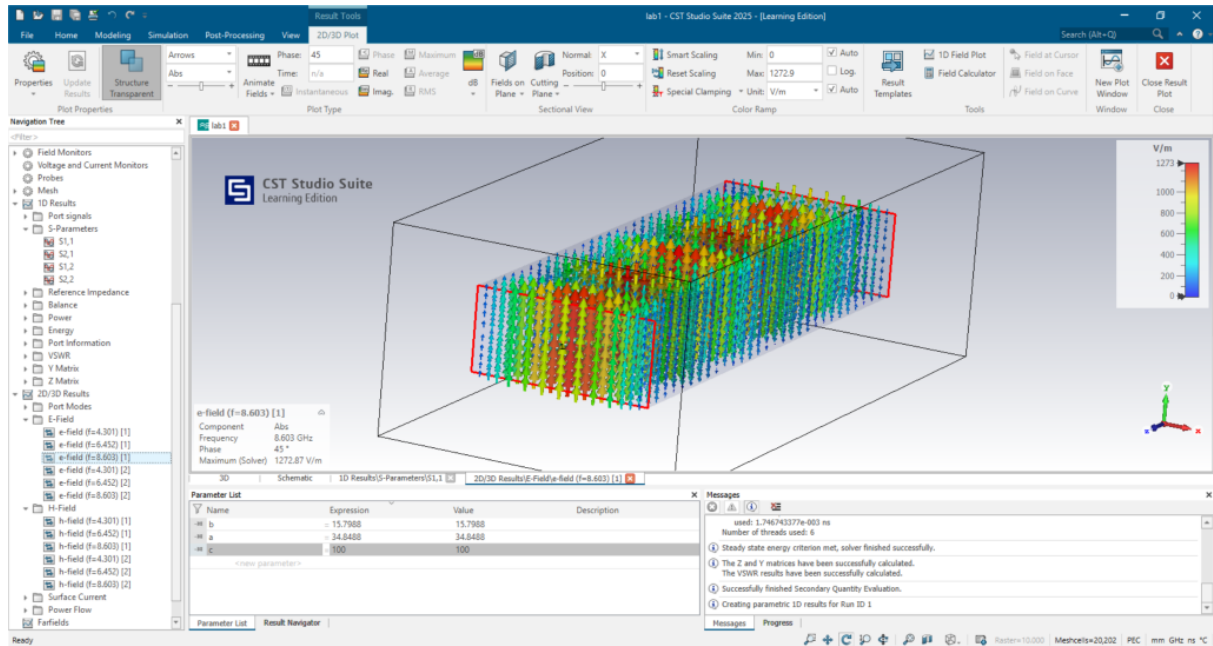


Fig 2.3 Electric field distribution for frequency = 8.603 GHz at Port 1

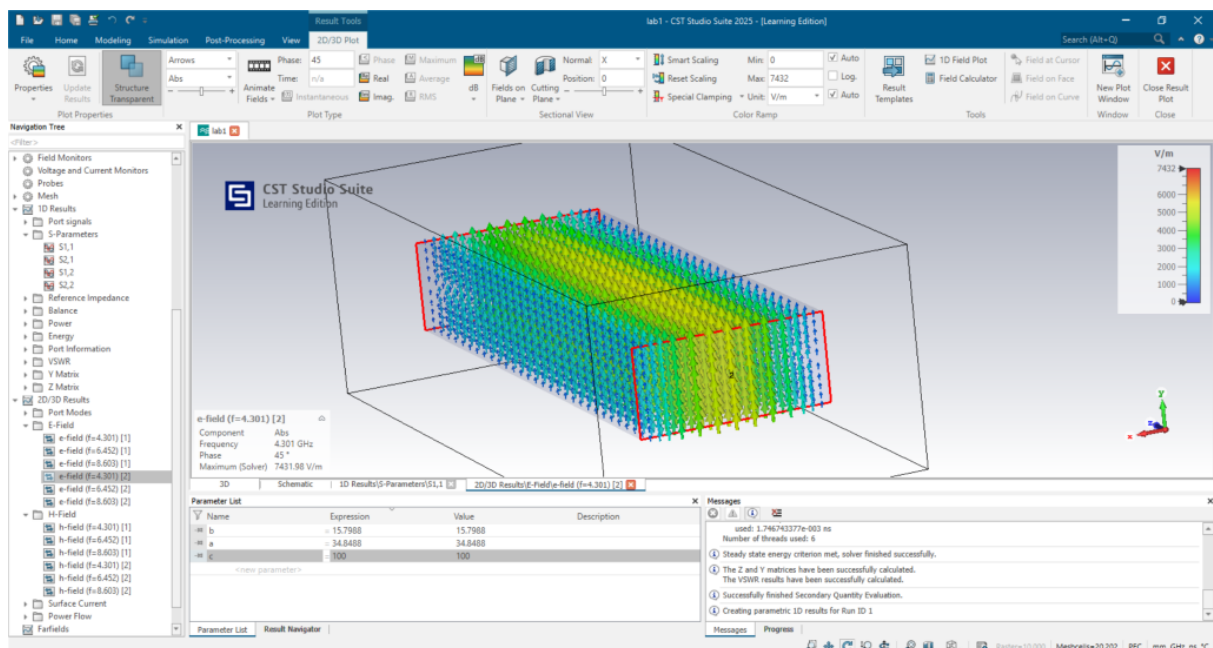


Fig 2.4 Electric field distribution for frequency = 4.301 GHz at Port 2

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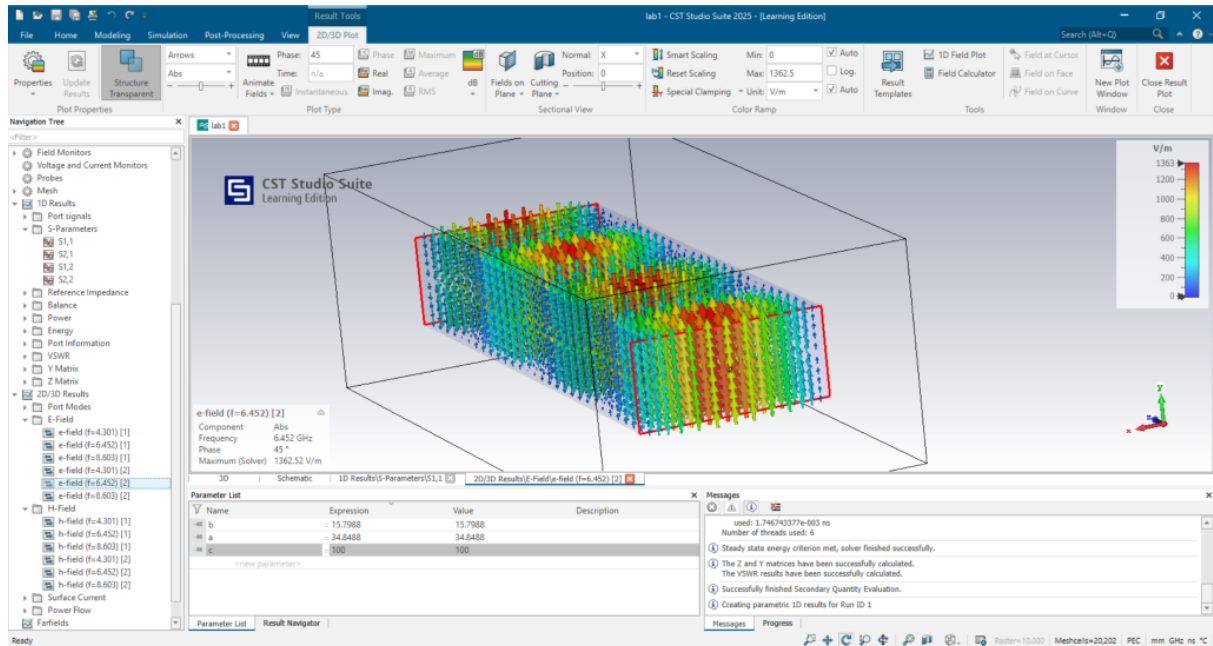


Fig 2.5 Electric field distribution for frequency = 6.452 GHz at Port 2

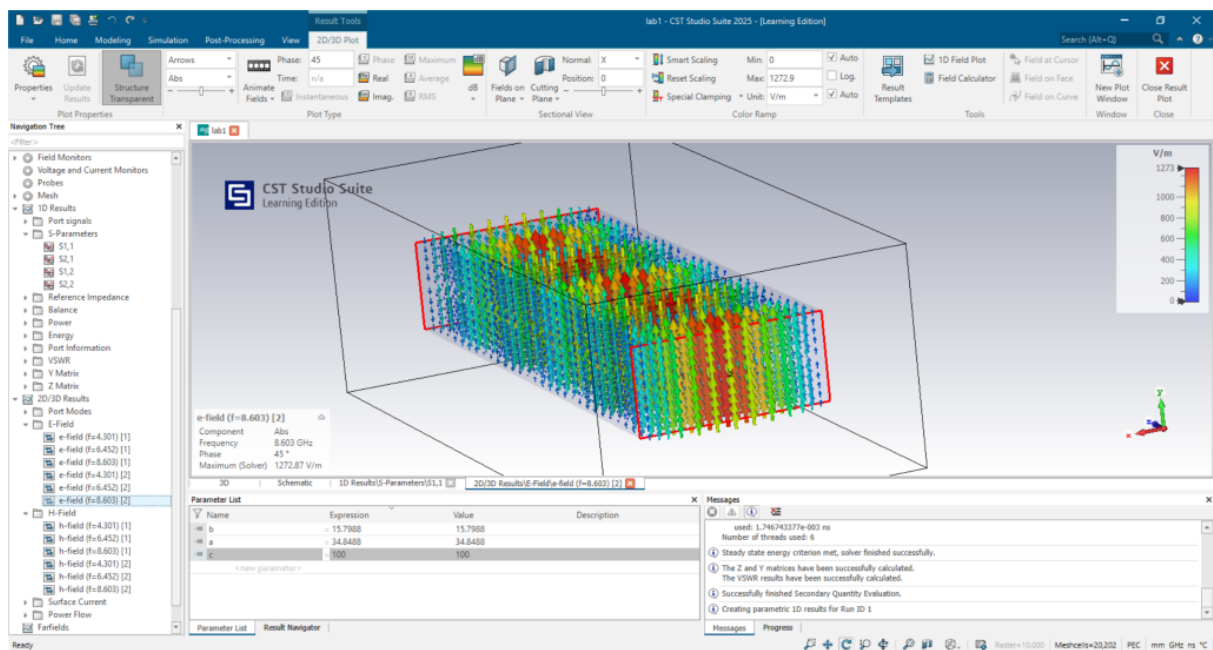


Fig 2.6 Electric field distribution for frequency = 8.603 GHz at Port 2

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Magnetic Field Distributions

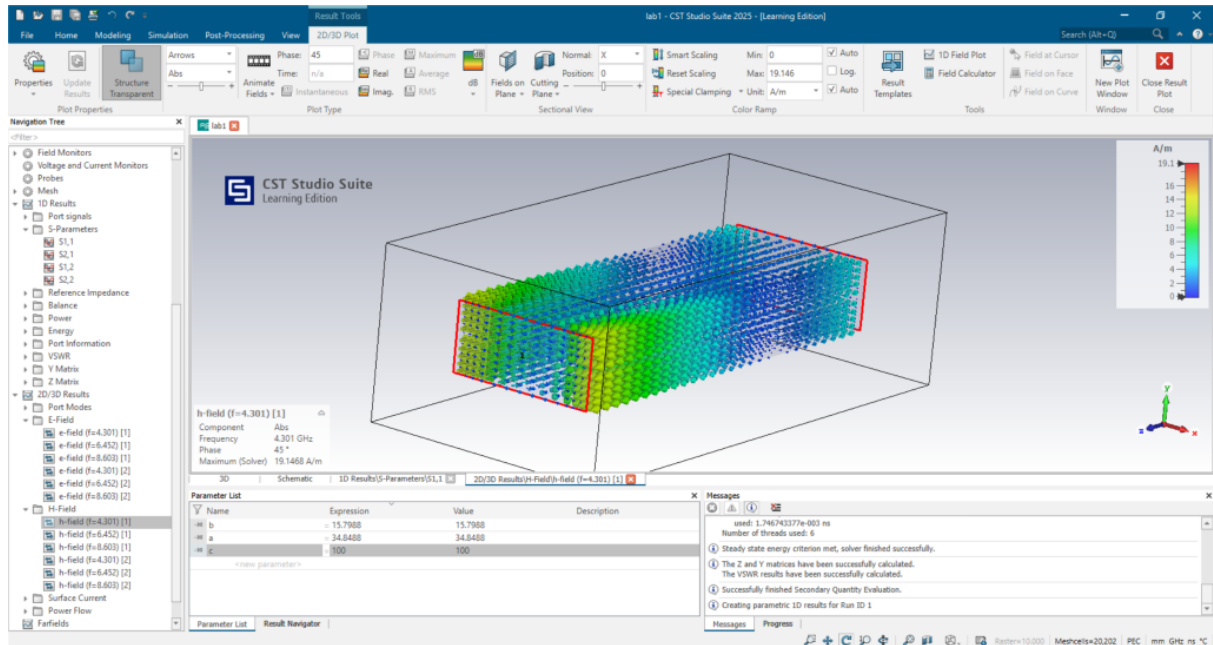


Fig 3.1 Magnetic field distribution for frequency = 4.301 GHz at Port 1

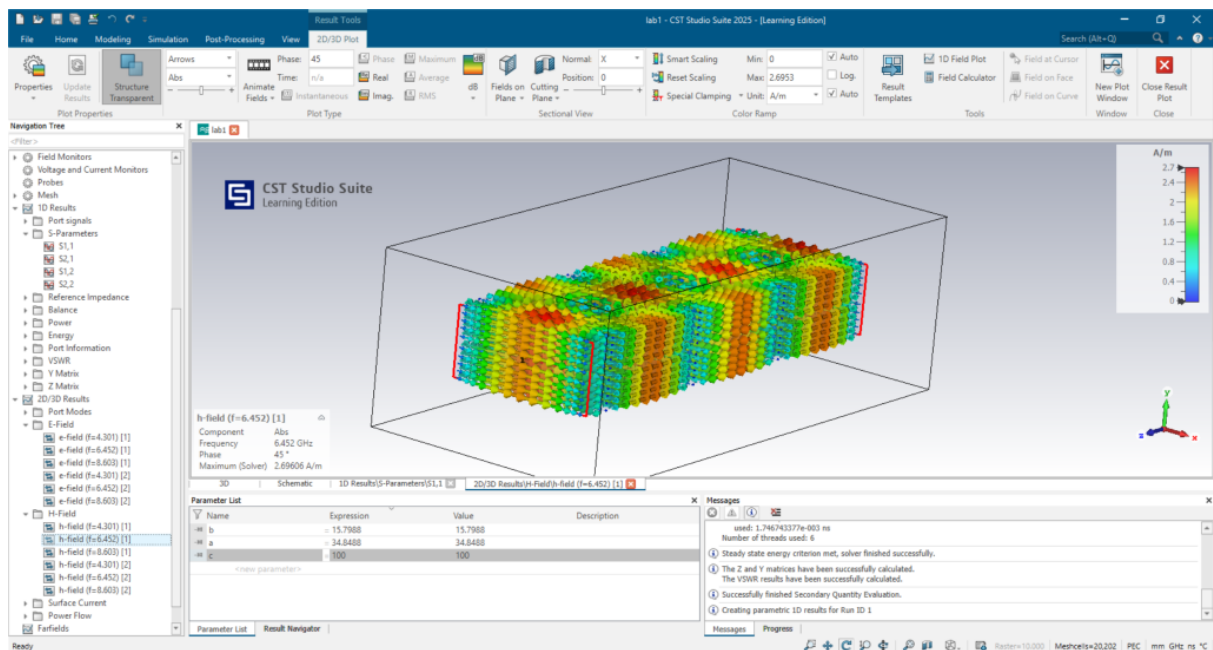


Fig 3.2 Magnetic field distribution for frequency = 6.452 GHz at Port 1

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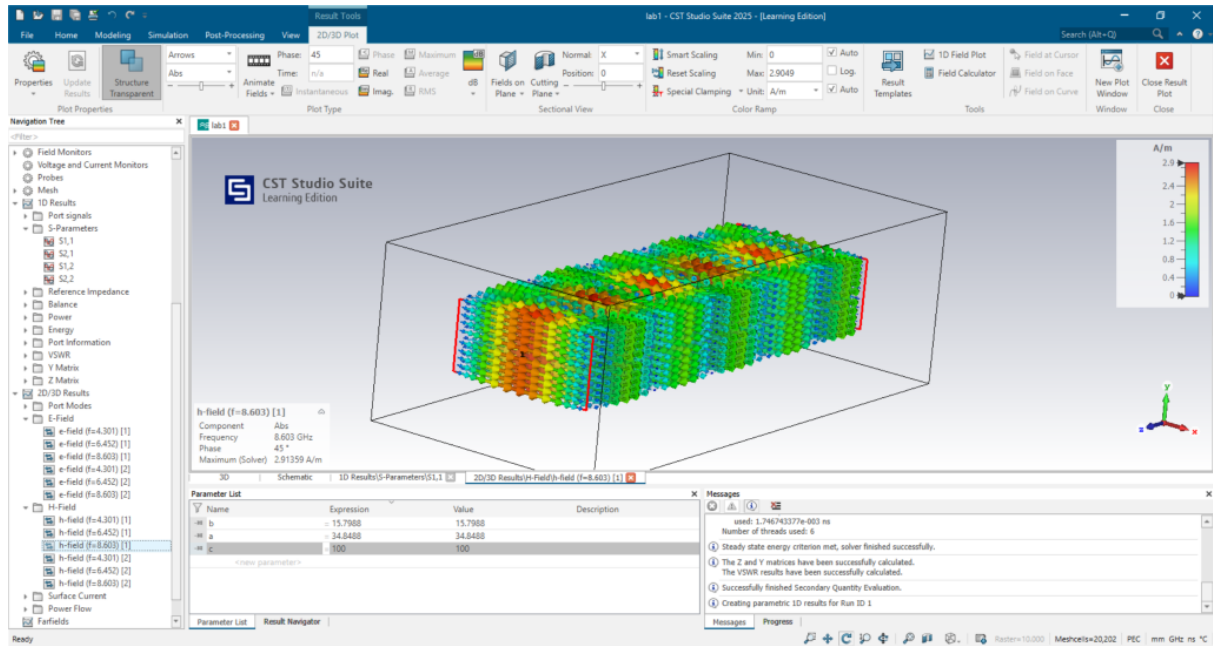


Fig 3.3 Magnetic field distribution for frequency = 8.603 GHz at Port 1

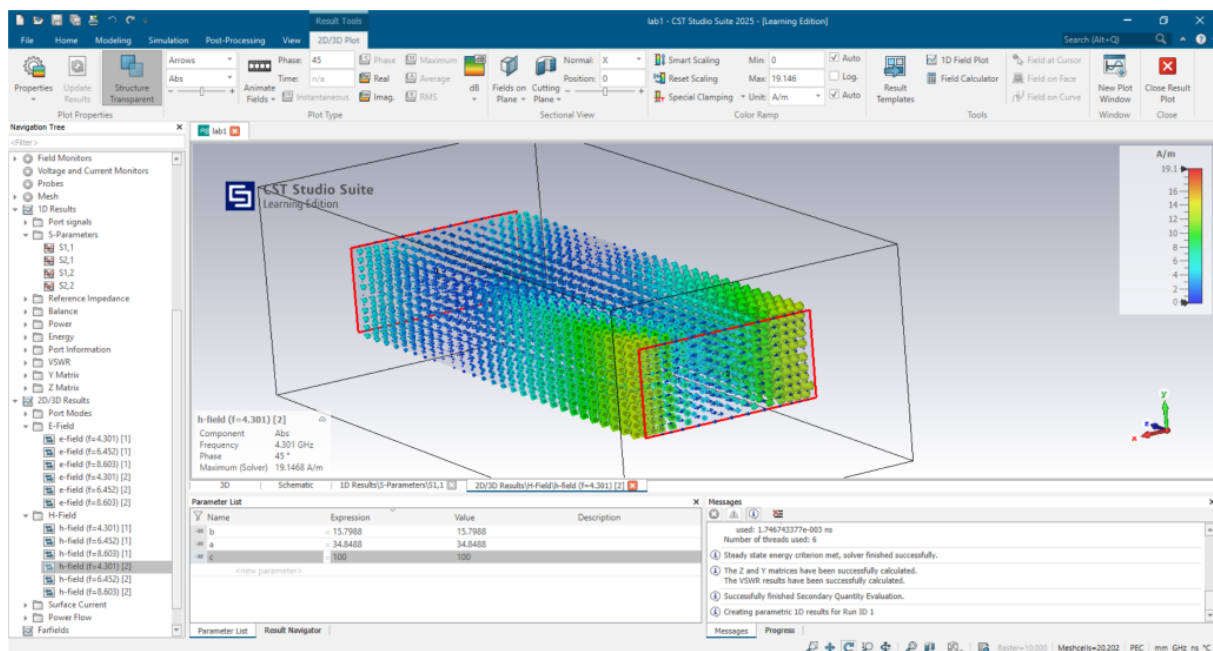


Fig 3.4 Magnetic field distribution for frequency = 4.301 GHz at Port 2

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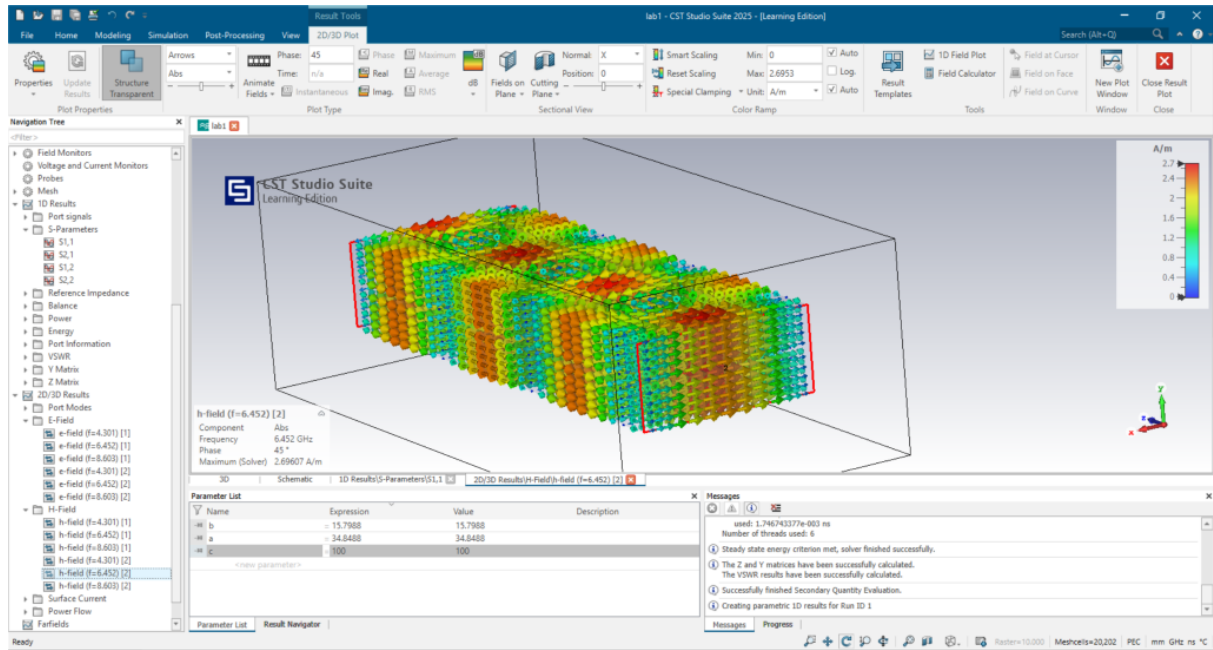


Fig 3.5 Magnetic field distribution for frequency = 6.452 GHz at Port 2

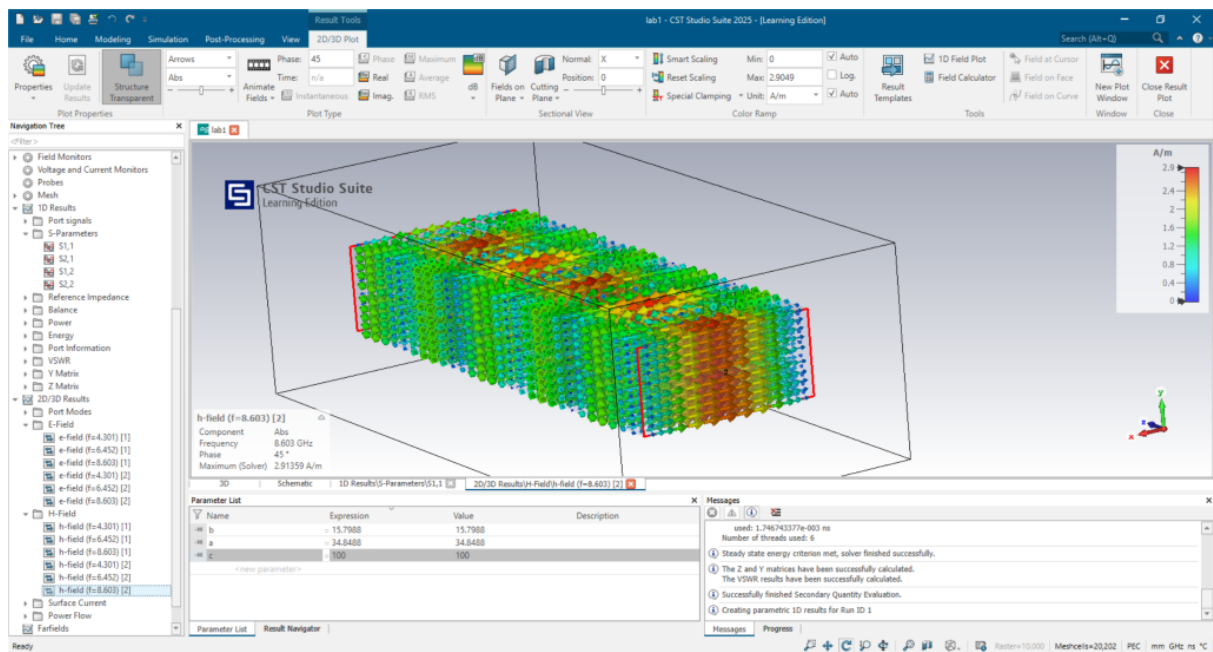


Fig 3.6 Magnetic field distribution for frequency = 8.603 GHz at Port 2

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Scattering Parameters

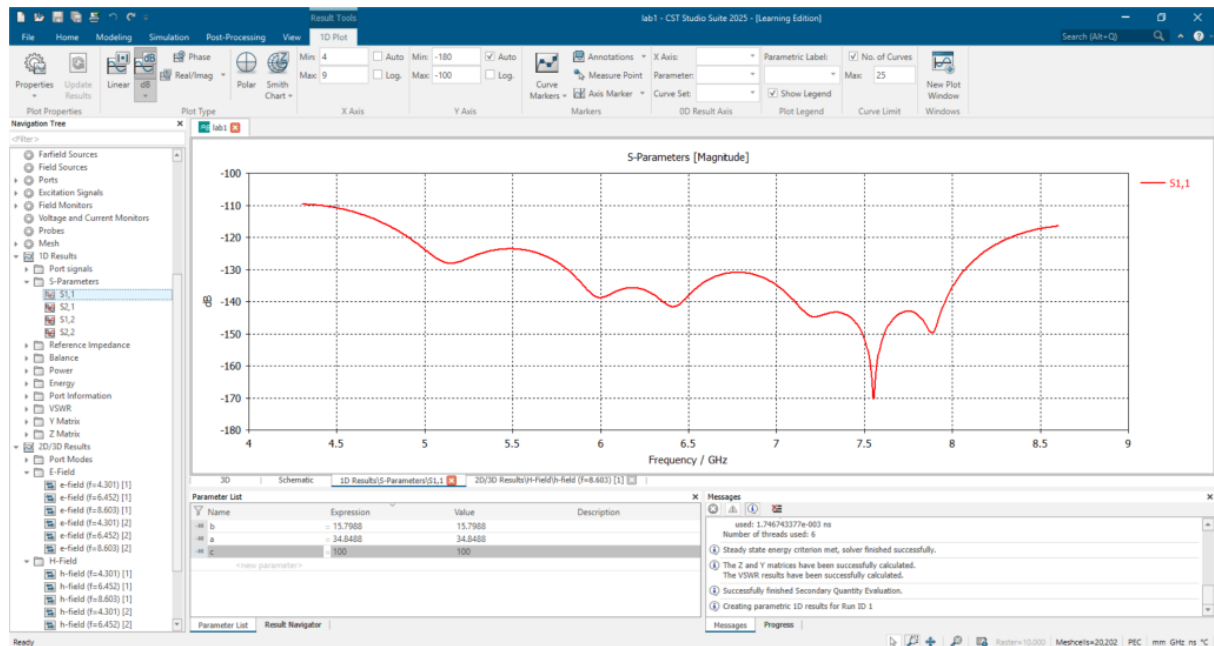


Fig 4.1 Scattering parameter $S(1,1)$

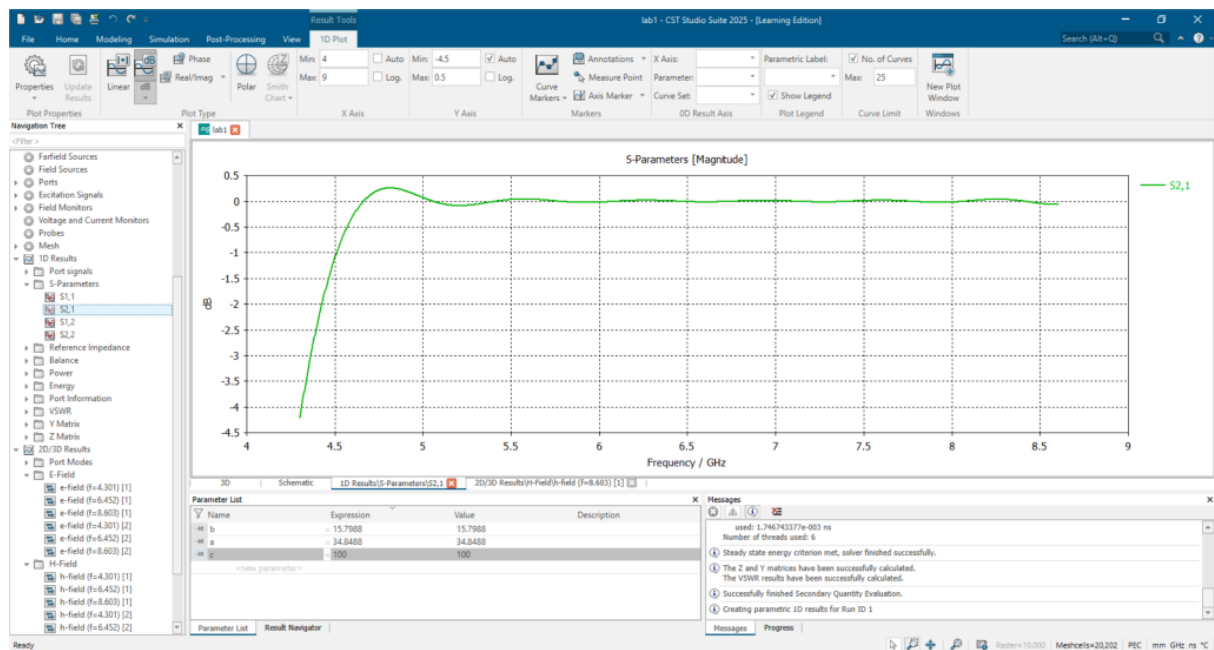


Fig 4.2 Scattering parameter $S(2,1)$

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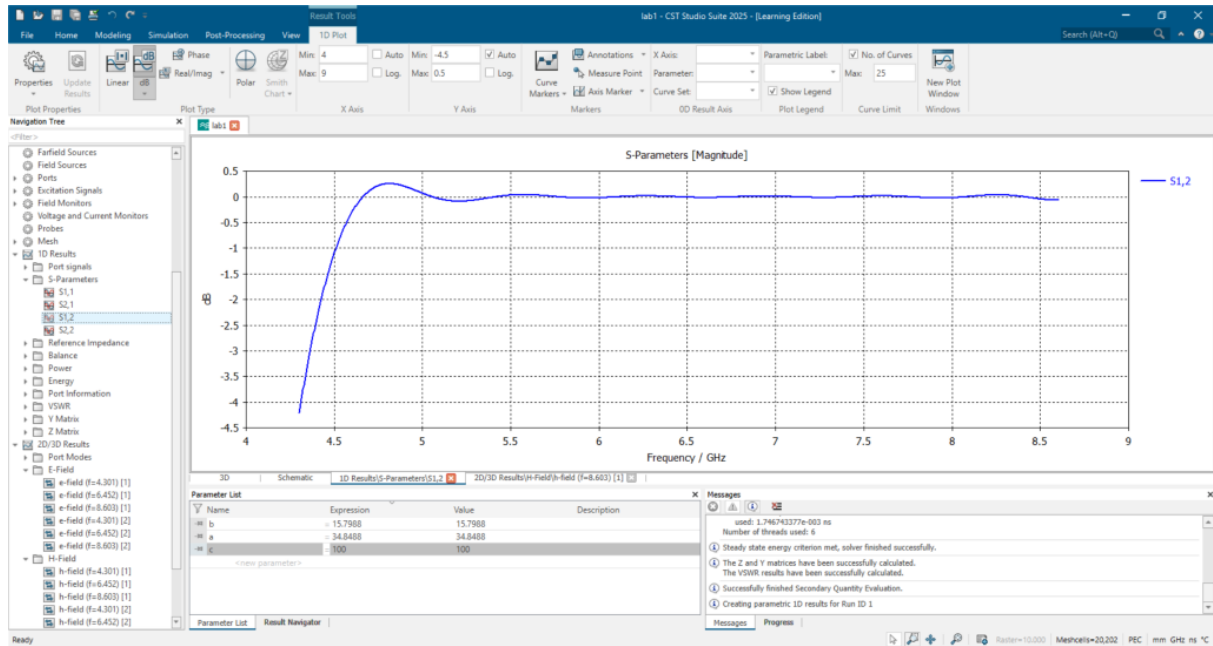


Fig 4.3 Scattering parameter S(1,2)

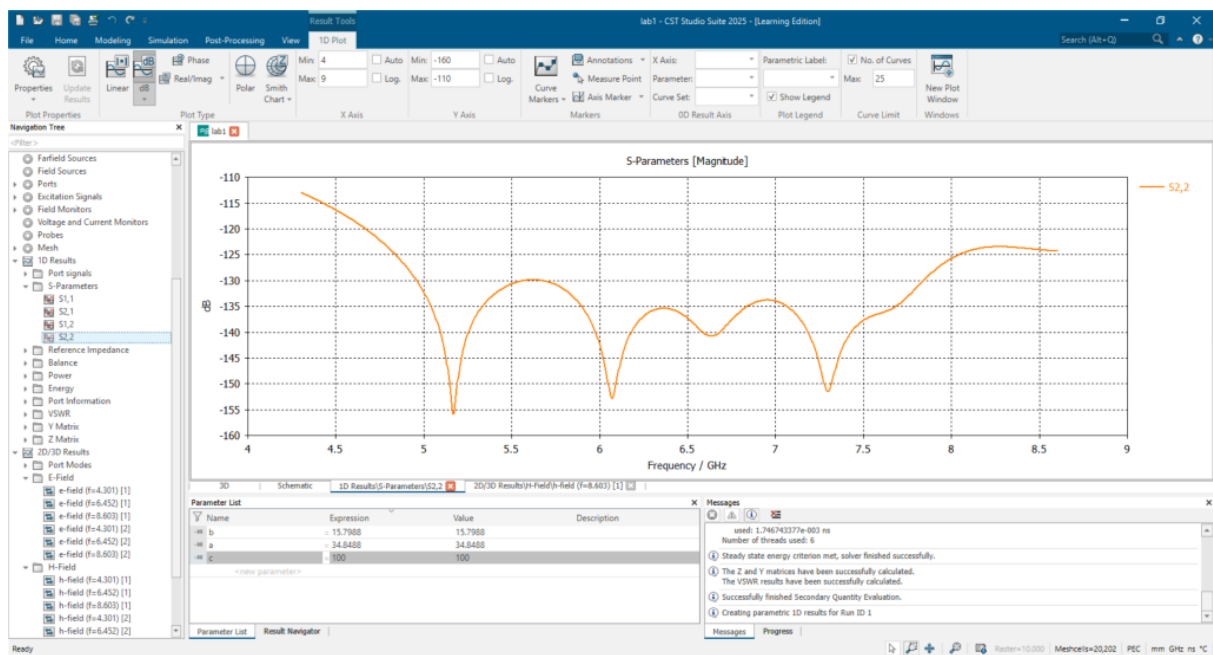


Fig 4.4 Scattering parameter S(2,2)

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Power Flow Distributions

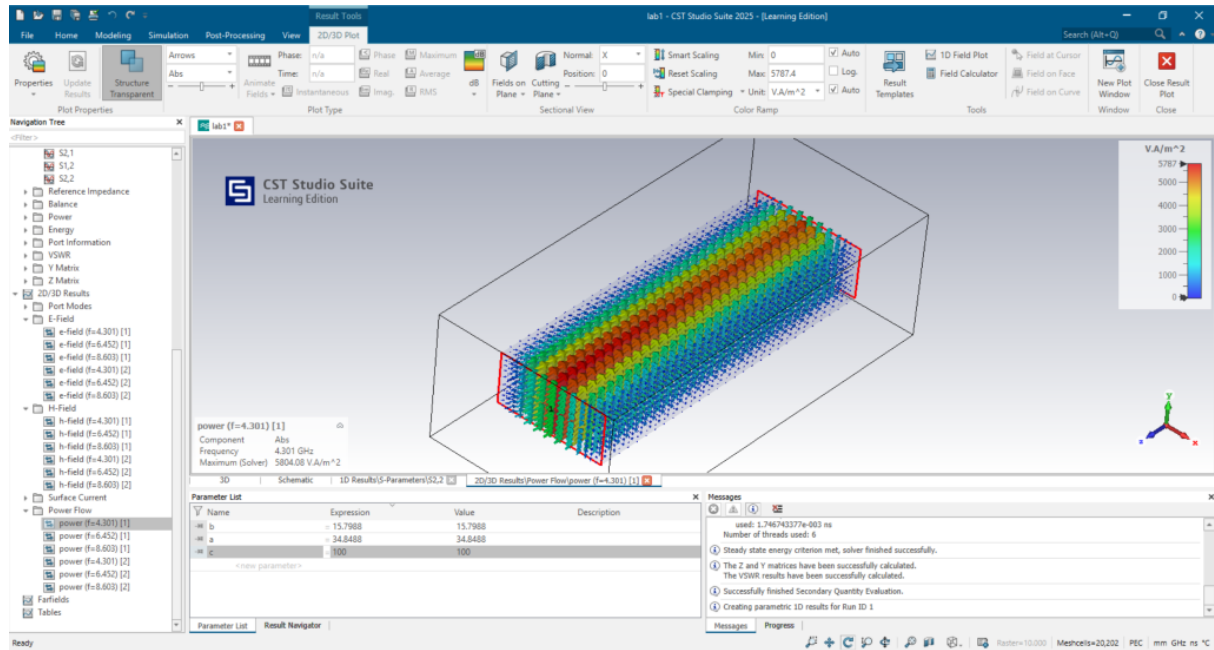


Fig 5.1 Power Flow for frequency = 4.301 GHz at Port 1

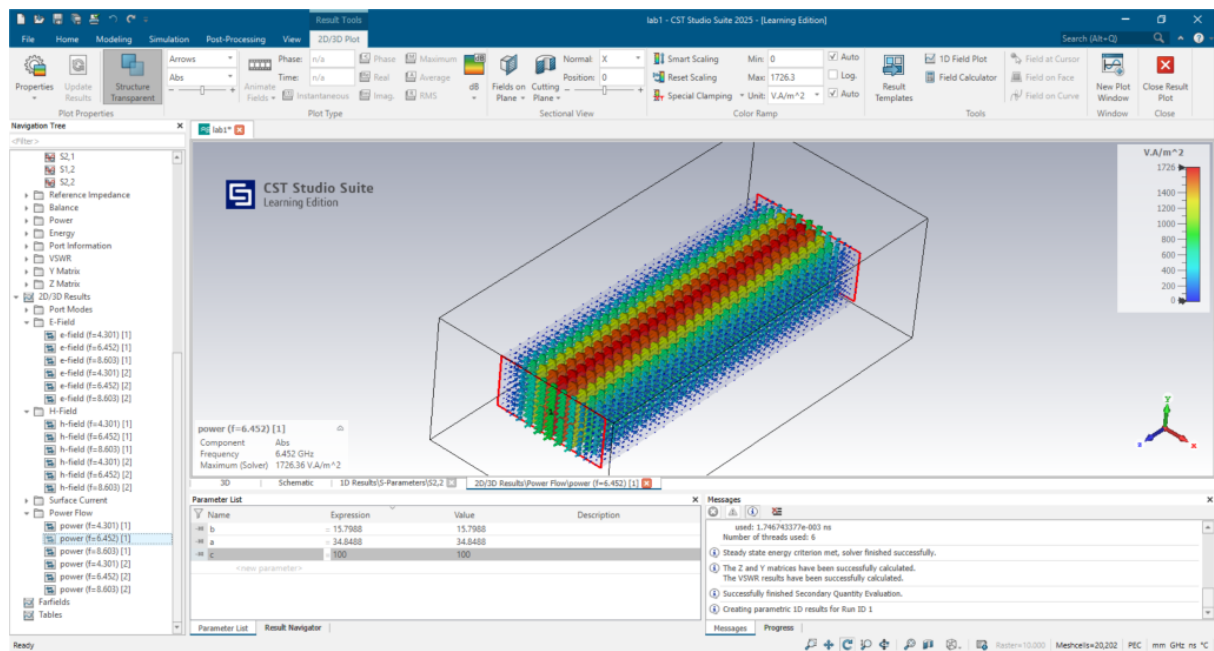


Fig 5.2 Power Flow for frequency = 6.452 GHz at Port 1

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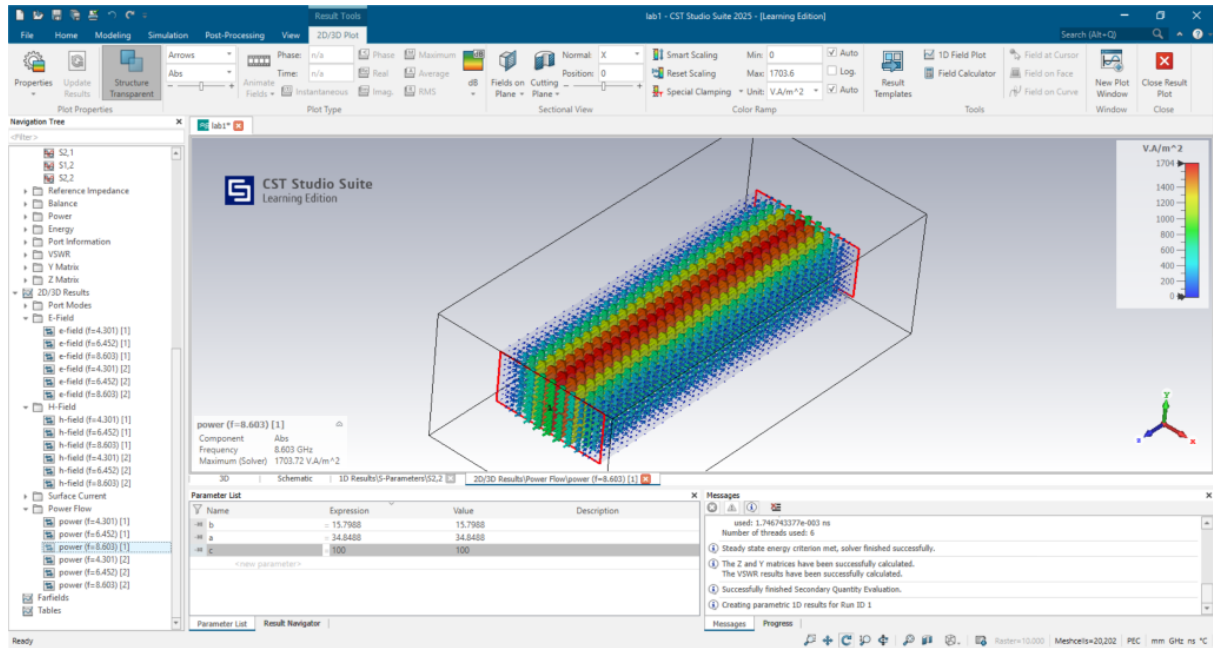


Fig 5.3 Power Flow for frequency = 8.603 GHz at Port 1

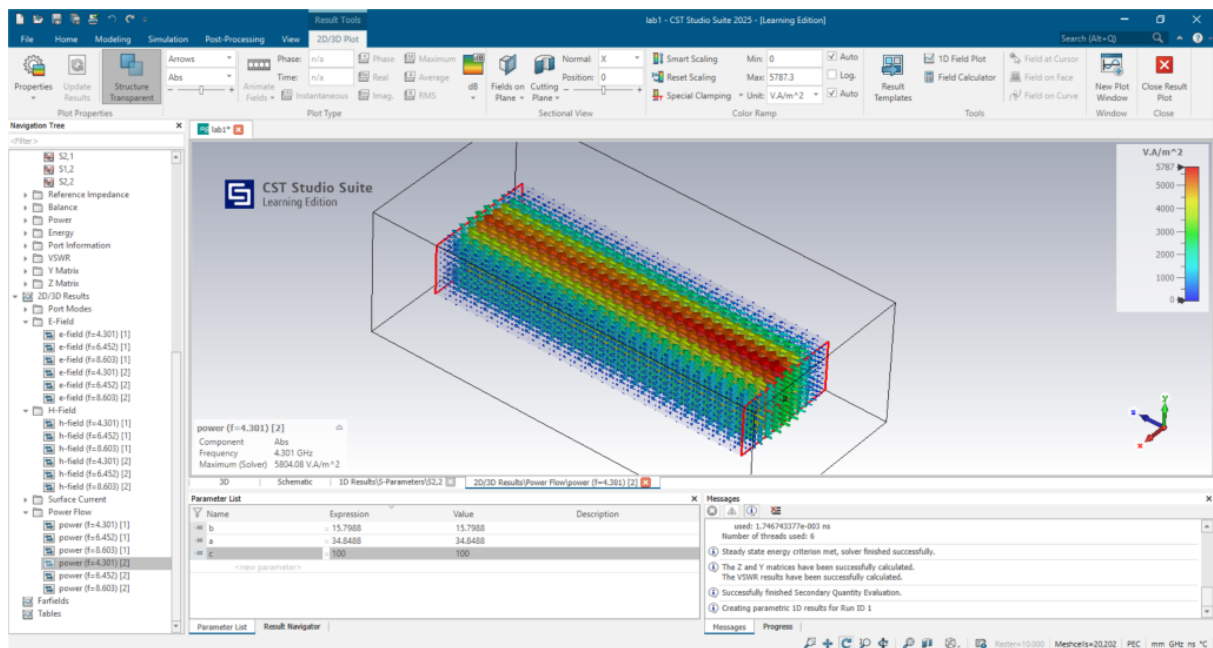


Fig 5.4 Power Flow for frequency = 4.301 GHz at Port 2

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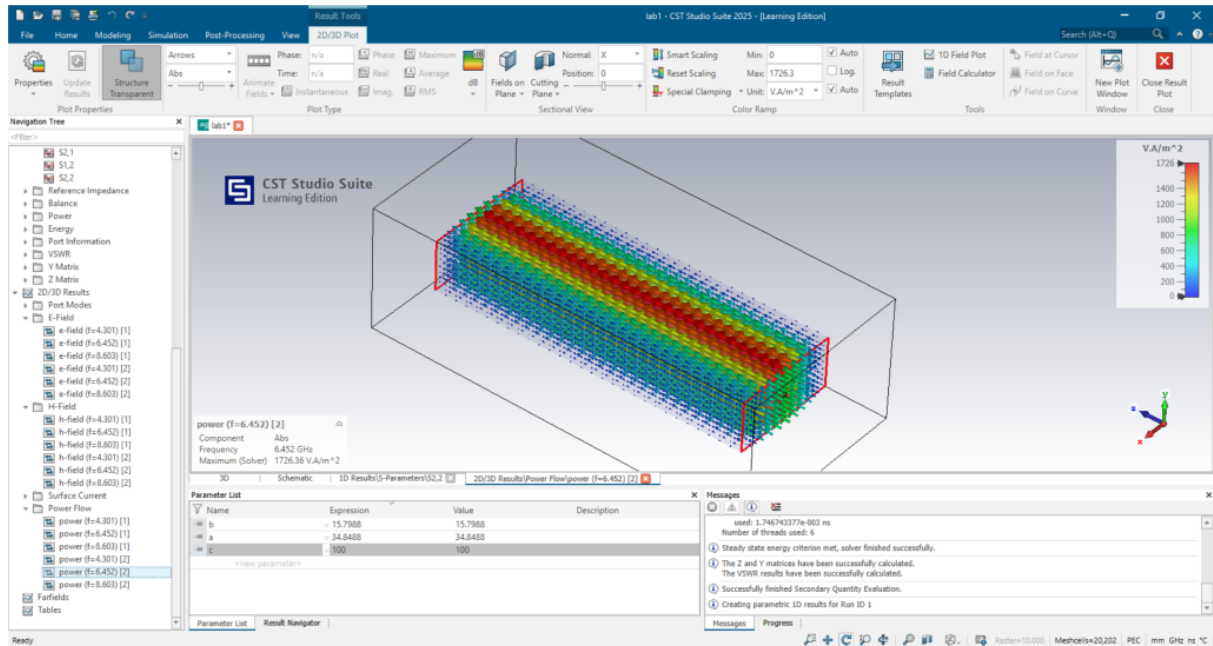


Fig 5.5 Power Flow for frequency = 6.452 GHz at Port 2

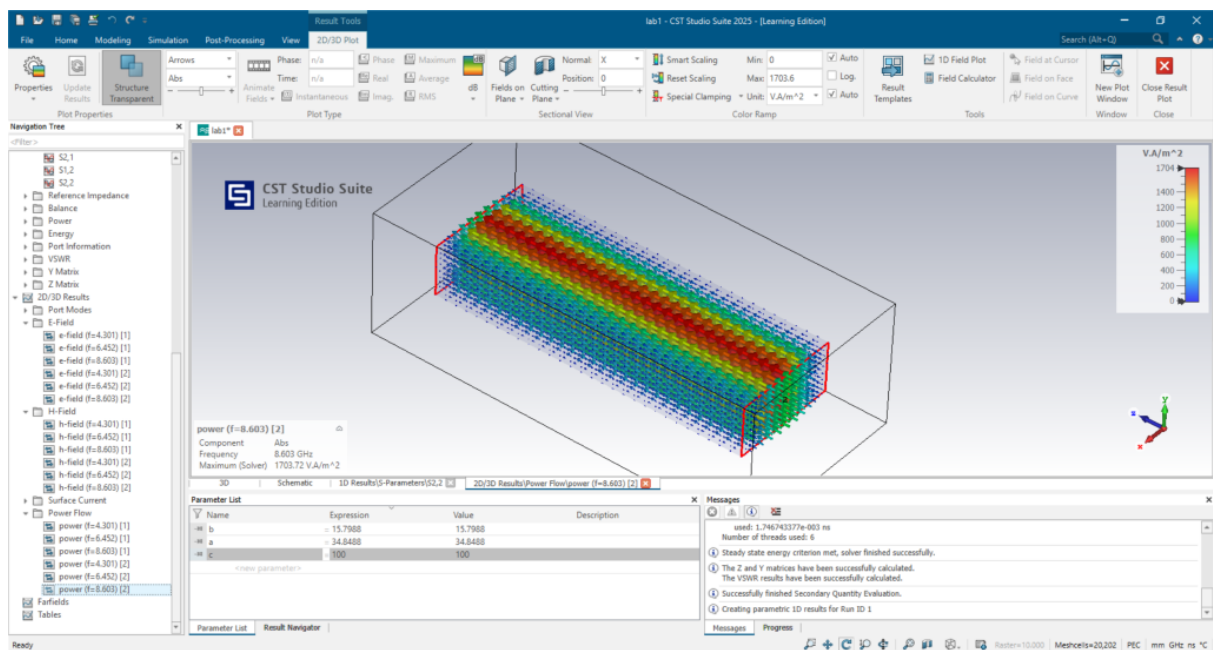


Fig 5.6 Power Flow for frequency = 8.603 GHz at Port 2

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Surface Current Distributions

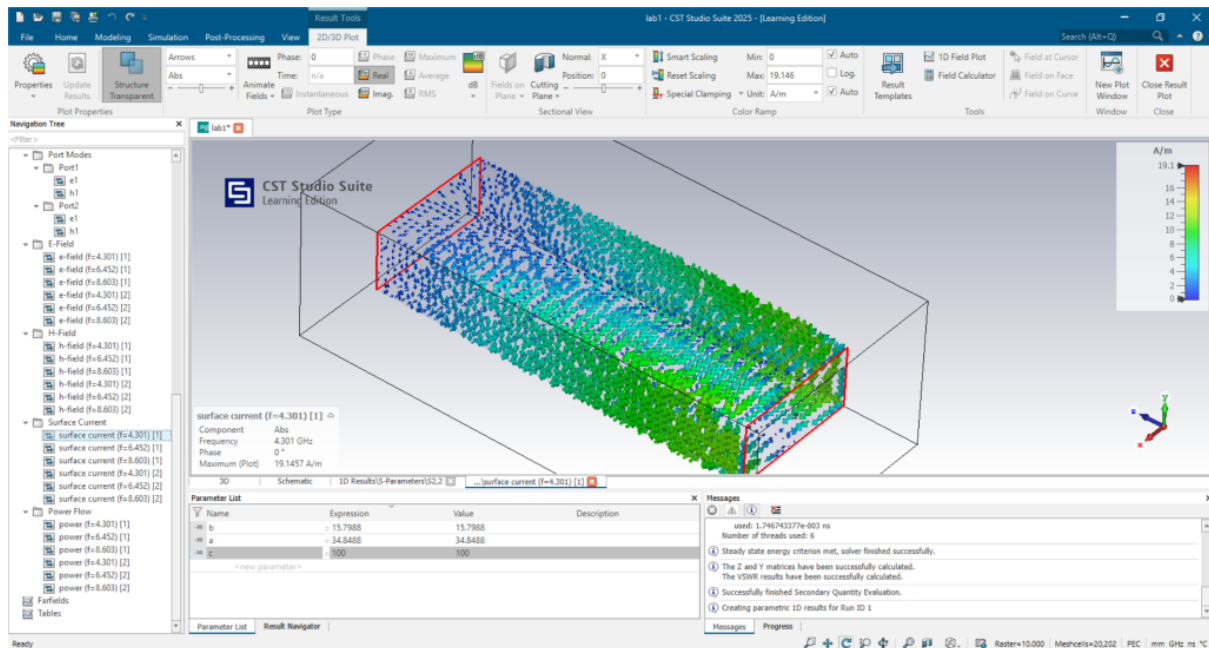


Fig 6.1 Surface Current Distribution for frequency = 4.301 GHz at Port 1

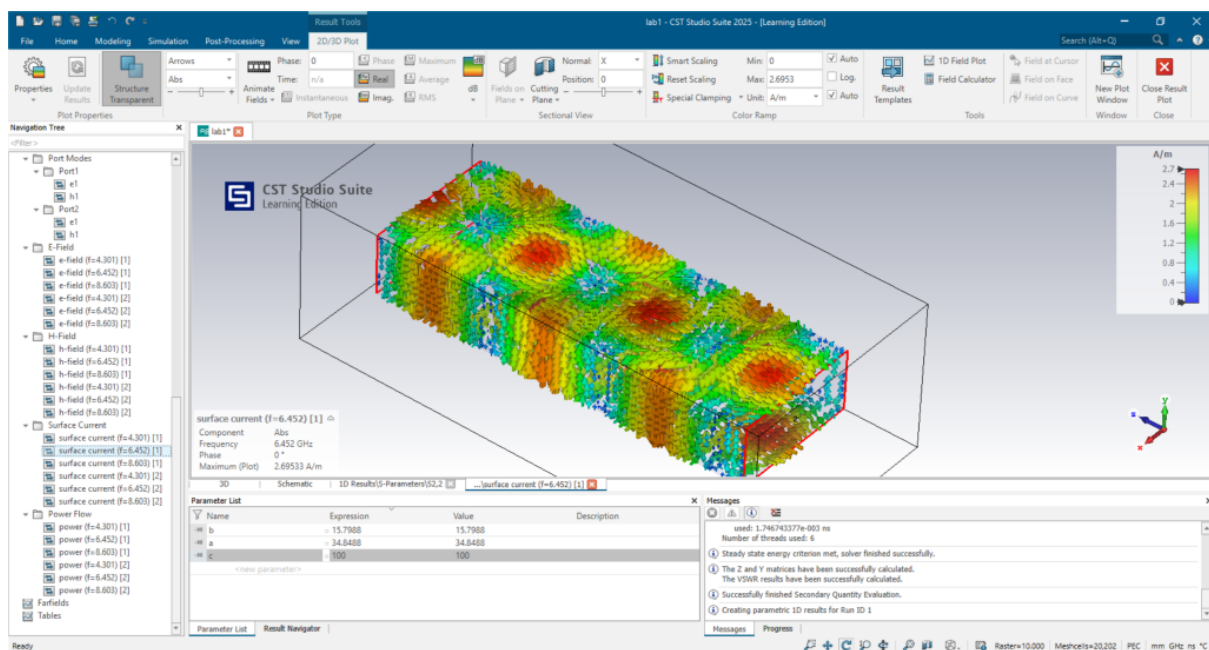


Fig 6.2 Surface Current Distribution for frequency = 6.452 GHz at Port 1

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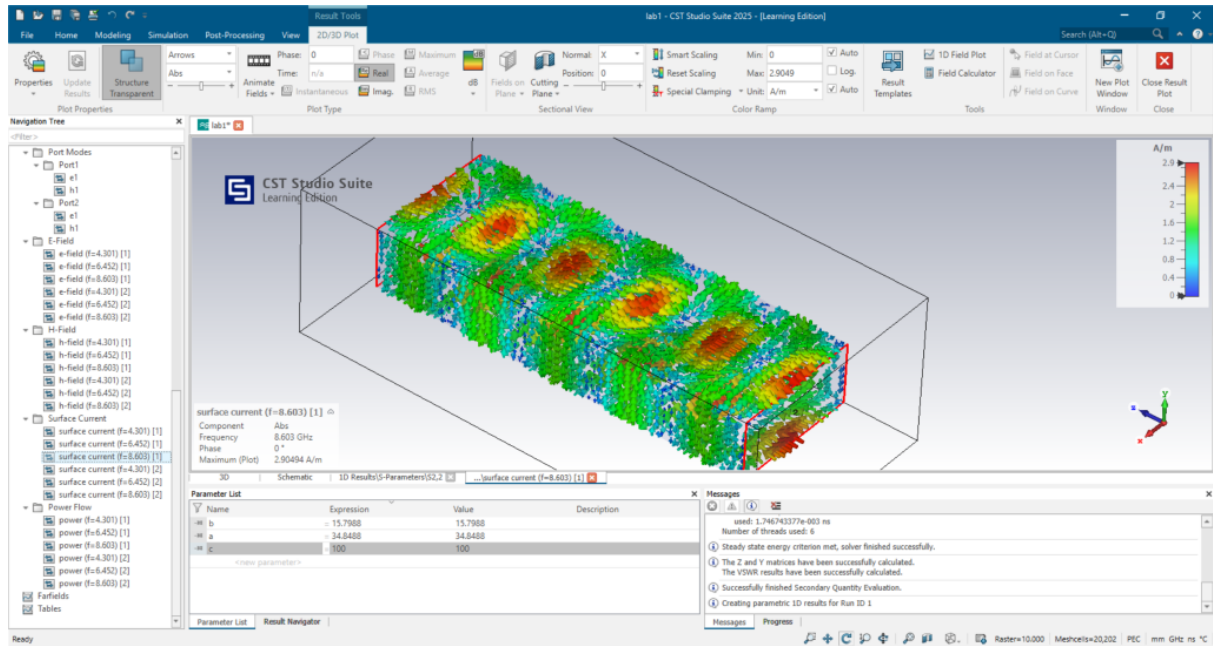


Fig 6.3 Surface Current Distribution for frequency = 8.603 GHz at Port 1

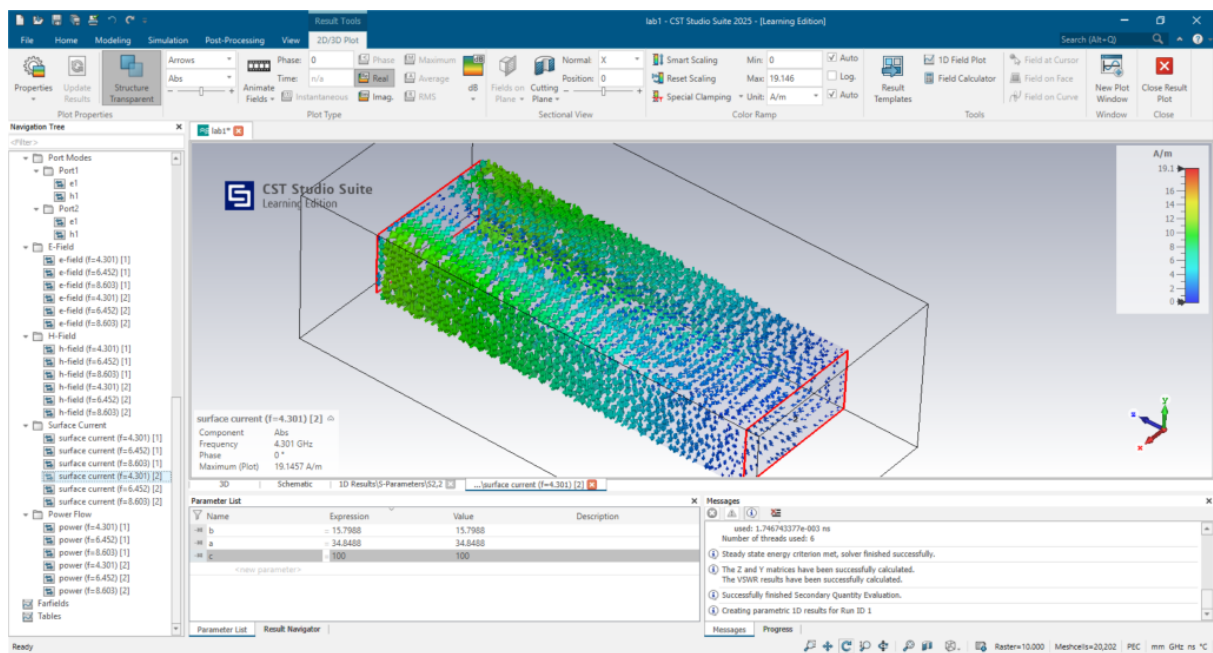


Fig 6.4 Surface Current Distribution for frequency = 4.301 GHz at Port 2

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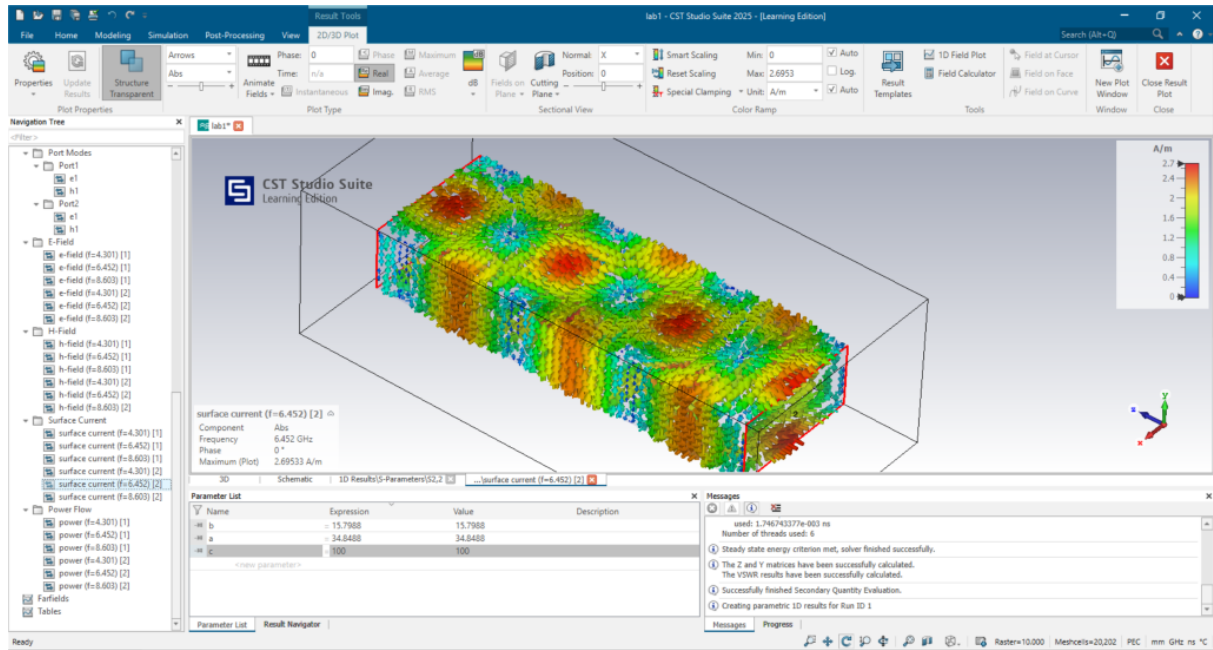


Fig 6.5 Surface Current Distribution for frequency = 6.452 GHz at Port 2

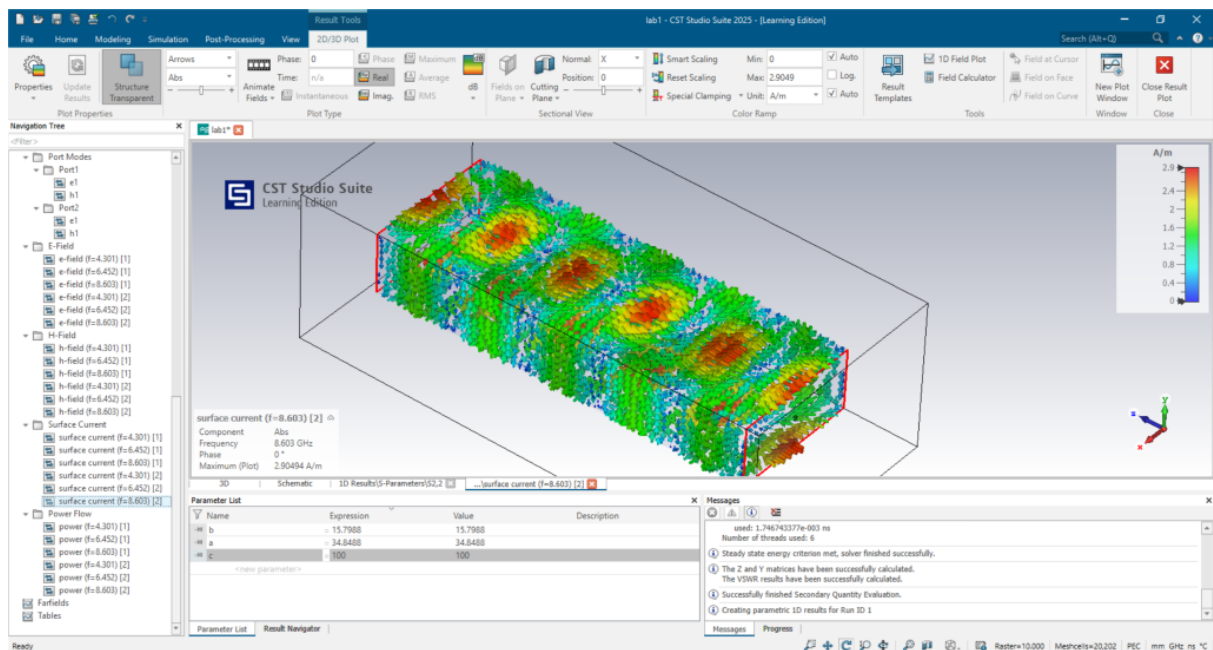


Fig 6.6 Surface Current Distribution for frequency = 8.603 GHz at Port 2

Modal Characteristics

The modal characteristics of the WR-137 waveguide were analyzed by verifying the field configurations and frequency response obtained from the simulation.

1. Verification of Dominant TE₁₀ Mode: The electric field distribution simulated at 6.452 GHz (Fig 2.2 and Fig 2.5) confirms the propagation of the dominant TE₁₀ mode.

- The electric field intensity exhibits a half-sine variation along the broad dimension (a), being maximum at the center and zero at the walls, which corresponds to the mode index $m=1$.
- The field remains constant along the narrow dimension (b), corresponding to the mode index $n=0$.
- The electric field vectors are observed to be parallel to the narrow dimension (y-axis), consistent with standard TE₁₀ topology.

2. Cutoff Frequency and Filtering Action: The scattering parameter results verify the high-pass filtering behavior of the waveguide.

- **Cutoff Frequency::** The transmission coefficient $S(21)$ plot (Fig 4.2) shows a sharp transition from high attenuation to transmission at approximately 4.3 GHz. This aligns with the theoretical cutoff frequency of 4.301 GHz for the WR-137 standard.
- **Evanescent vs. Propagating Modes:** Below 4.3 GHz, the $S(21)$ value is low (Fig 4.2) and the field distributions (Fig 3.1 and Fig 3.4) show weak or non-propagating energy. Above 4.3 GHz (Fig 4.2), $S(21)$ approaches 0 dB, indicating successful propagation.

3. Magnetic Field Characteristics The magnetic field distributions (Fig 3.2 and Fig 3.5) display characteristic loops in the x-z plane. The presence of a longitudinal magnetic field component (H_z) and the absence of a longitudinal electric field component (E_z) conclusively verify the Transverse Electric (TE) nature of the mode.

Conclusions:

The electromagnetic analysis of the standard WR-137 rectangular waveguide was successfully completed using CST Studio Suite. The simulation results conclusively verified the theoretical principles of waveguide propagation and provided the following key insights:

1. **Modal Characteristics:** The waveguide demonstrated clear high-pass filtering behavior. The simulated cutoff frequency was observed at approximately **4.3 GHz**, which aligns with the theoretical value calculated for the WR-137 standard. This confirms that the structure supports propagation only above this threshold.
2. **Field Distributions:** The Electric field distributions visualized at operating frequencies (e.g., 6.452 GHz) exhibited a half-sine variation along the broad dimension and a constant profile along the narrow dimension. This specific field configuration, along with the magnetic field loops observed in the x-z plane, confirms the propagation of the dominant **TE₁₀ mode**.
3. **Scattering Parameters (S-Parameters):** The S-parameter analysis showed excellent transmission characteristics within the passband. The Transmission Coefficient S_{21} approached 0 dB above the cutoff frequency, indicating minimal signal loss. Conversely, the Reflection Coefficient S_{11} remained low in the operating band, signifying good impedance matching and efficient energy transfer.
4. **Power Flow:** The Poynting vector visualizations demonstrated that electromagnetic power is effectively confined within the metallic walls and flows unidirectionally from the input port to the output port without leakage, validating the waveguide's efficiency as a transmission medium.
5. **Surface Current Distributions:** The simulation revealed significant surface current densities on the broad walls of the waveguide, with current lines running parallel to the direction of propagation at the center and diverging towards the edges. These distributions satisfy the boundary conditions required for TE mode propagation in a rectangular conductor.

In summary, the design and simulation of the WR-137 waveguide have been validated, with all simulated performance metrics, including cutoff frequency, modal patterns, and scattering parameters.